Riemannian Geometry and Geometric Field Equations of Two-Agent Interaction Dynamics: From Time-Dependent to State-Dependent Manifolds

Chewin Pinmook

Contact: chewinp@gmail.com

May 2026

Abstract

This work presents a unified geometric framework for two-agent interaction dynamics, developed in two successive stages. In the first stage, we construct a time-dependent Riemannian metric on the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ by imposing a linear constraint $I_P = \gamma(t) I_S$ on the passive interaction variable. The induced metric $G(t) = I + \gamma^T \gamma$ gives rise to a natural interaction flow tensor

$$K^i{}_j = \tfrac{1}{2} G^{ik} \dot{G}_{kj},$$

whose time evolution governs all Ricci curvature components. In the second stage, we extend this model to a state-dependent framework by allowing $\gamma = \gamma(t, I_S)$, which introduces spatial variation in the interaction metric

$$G(t, I_S) = I + J^T J, \quad J = \gamma + (\nabla_{I_S} \gamma) I_S.$$

This extension generates non-trivial spatial curvature and leads naturally to a Ricci tensor decomposition into temporal and spatial contributions. Building on this geometric foundation, we introduce an interaction action functional and prove, under the axioms of locality, diffeomorphism covariance, and second-order field equations, that the admissible action is uniquely determined by the scalar curvature. Variation yields the interaction field equation

$$R_{\mu\nu} - \tfrac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

which is structurally analogous to the Einstein field equations. The framework provides a rigorous nonlinear description of interaction dynamics, where $T_{\mu\nu}$ encodes reinforcement and perceptual structures, offering a unified perspective linking differential geometry with behavioral feedback systems.

Contents

1 Introduction

Mathematical descriptions of interacting systems often rely on dynamical equations defined in Euclidean state spaces. However, when interaction variables mutually constrain one another, the underlying state space may naturally acquire a nontrivial geometric structure. In such cases, curvature provides a coordinate-invariant measure of structural deformation induced by temporal evolution.

This work develops the geometric theory of two-agent interaction dynamics in two successive stages, motivated by increasing levels of physical realism.

In the *first stage* (Sections 2–3), we consider a time-only interaction rule $\gamma = \gamma(t)$, where a passive interaction variable I_P is linearly constrained by an active variable I_S through a time-dependent matrix transformation. This yields a block-diagonal Riemannian metric whose spatial components evolve purely in time. All curvature then arises from temporal deformation, and the system is equivalent to a cosmological-type metric evolution.

In the *second stage* (Sections 4–6), we generalise to $\gamma = \gamma(t, I_S)$, so that the interaction rule depends on the current interaction state itself. This extension is motivated by many real-world phenomena:

- nonlinear feedback in dynamical systems,
- emotional escalation and saturation in social interactions,
- biological response saturation,
- adaptive feedback control.

The state dependence introduces spatial variation in the metric, generating non-trivial intrinsic curvature even in the absence of temporal deformation. We then formulate a variational principle and derive Einstein-type interaction field equations from first principles.

Throughout, the key quantities are:

- the *interaction flow tensor* $K^i{}_j$, governing temporal metric evolution,
- the *emotional quantity tensor* E_{ij}, measuring intrinsic interaction acceleration,
- the *interaction source tensor* $T_{\mu\nu}$, encoding reinforcement and perceptual forcing.

2 Stage I: Time-Dependent Interaction Metric

2.1 Interaction Variables and Constraint

We consider two interaction vectors

$$I_S(t) \in \mathbb{R}^2, \qquad I_P(t) \in \mathbb{R}^2,$$

representing active and passive interaction components, respectively. We impose the *linear interaction constraint*

$$I_P(t) = \gamma(t)\, I_S(t),$$

where $\gamma(t) \in \mathbb{R}^{2\times 2}$ is a differentiable matrix-valued function of time.

2.2 Extended Euclidean Construction

Consider the extended coordinate space

$$(t,\, I_P^1,\, I_P^2,\, I_S^1,\, I_S^2) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2,$$

equipped with the standard Euclidean line element

$$ds^2 = dt^2 + dI_P^T\, dI_P + dI_S^T\, dI_S.$$

2.3 Induced Interaction Metric

Imposing the constraint gives $dI_P = \gamma(t)\, dI_S$ at fixed t. Substituting into the Euclidean line element:

$$ds^2 = dt^2 + dI_S^T \big(I + \gamma(t)^T \gamma(t)\big)\, dI_S.$$

We therefore define the *spatial interaction metric*

$$G(t) = I + \gamma(t)^T \gamma(t).$$

By construction $G(t)$ is symmetric, positive definite, and depends only on time. The full metric on the manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ is

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{pmatrix},$$

where $a > 0$ is a constant temporal scale factor.

3 Stage I: Curvature from Temporal Deformation

Definition 3.1 (Interaction Flow Tensor)**.** *The* interaction flow tensor *is*

$$K^i{}_j = \tfrac{1}{2}\, G^{ik}\dot{G}_{kj},$$

where $\dot{G}_{kj} = dG_{kj}/dt$. This tensor measures the rate of temporal deformation of the interaction metric.

Theorem 3.1 (Non-vanishing Christoffel Symbols)**.** *For the metric above, the only non-vanishing Christoffel symbols are*

$$\Gamma^0_{ij} = -\frac{1}{2a^2}\dot{G}_{ij}, \qquad \Gamma^i_{0j} = \Gamma^i_{j0} = K^i{}_j.$$

Proof. From $\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$ and the block-diagonal structure with $g_{0i} = 0$, $g_{00} = a^2$ constant, and $\partial_k G_{ij} = 0$, the only surviving terms are those involving $\partial_t G_{ij}$. □

Theorem 3.2 (Ricci Tensor: Temporal Component)**.**

$$R_{00} = -\operatorname{Tr}(\dot{K}) - \operatorname{Tr}(K^2).$$

Proof. Since $\Gamma^\lambda_{00} = 0$, the first and third terms in the Ricci formula vanish. The second term gives $-\partial_t\Gamma^i_{0i} = -\operatorname{Tr}(\dot{K})$, and the quartic contraction yields $-\operatorname{Tr}(K^2)$. □

Definition 3.2 (Emotional Quantity Tensor)**.** *The* emotional quantity tensor *is*

$$E_{ij} = \dot{K}_{ij} + K_{ik}K^k{}_j.$$

It describes the intrinsic interaction acceleration independent of collective coupling.

Theorem 3.3 (Ricci Tensor: Spatial Components)**.**

$$R_{ij} = E_{ij} + K_{ij}\operatorname{Tr}(K) = -\frac{1}{a^2}\left[\dot{K}_{ij} + \operatorname{Tr}(K)K_{ij} - 2K_{ik}K^k_j\right].$$

Proof. We evaluate

$$R_{ij} = \partial_\alpha\Gamma^\alpha_{ij} - \partial_j\Gamma^\alpha_{i\alpha} + \Gamma^\alpha_{\alpha\beta}\Gamma^\beta_{ij} - \Gamma^\alpha_{j\beta}\Gamma^\beta_{\alpha i}$$

term by term, using the connection of Theorem 3.1.

First term. The only nonvanishing Γ^α_{ij} is $\Gamma^0_{ij} = -\frac{1}{2a^2}\dot{G}_{ij} = -\frac{1}{a^2}K_{ij}$. Hence $\partial_\alpha\Gamma^\alpha_{ij} = \partial_0\Gamma^0_{ij} = -\frac{1}{a^2}\dot{K}_{ij}$.

Second term. $\Gamma^\alpha_{i\alpha} = \Gamma^0_{i0} + \Gamma^k_{ik} = 0$ identically in Stage I (the first vanishes since g_{00} is constant, the second since $\partial_k G_{ij} = 0$). So $\partial_j\Gamma^\alpha_{i\alpha} = 0$.

Third term. Only $\beta = 0$ contributes, since $\Gamma^{\beta}_{ij} \neq 0$ only for $\beta = 0$. With $\Gamma^{\alpha}_{\alpha 0} = \Gamma^{0}_{00} + \Gamma^{k}_{k0} = \mathrm{Tr}(K)$, this term equals $-\frac{1}{a^2}\mathrm{Tr}(K)\, K_{ij}$.

Fourth term. The only nonvanishing Christoffel symbols with a mixed 0/spatial index structure are $\Gamma^{0}_{ij} = -K_{ij}/a^2$ and $\Gamma^{i}_{0j} = K^{i}{}_{j}$. The sum $\Gamma^{\alpha}_{j\beta}\Gamma^{\beta}_{\alpha i}$ receives two equal contributions, from $(\alpha, \beta) = (0, k)$ and $(\alpha, \beta) = (k, 0)$:

$$\Gamma^{0}_{jk}\Gamma^{k}_{0i} + \Gamma^{k}_{j0}\Gamma^{0}_{ki} = -\tfrac{1}{a^2}K_{jk}K^{k}{}_{i} - \tfrac{1}{a^2}K^{k}{}_{j}K_{ik} = -\tfrac{2}{a^2}K_{ik}K^{k}{}_{j},$$

so that $-\Gamma^{\alpha}_{j\beta}\Gamma^{\beta}_{\alpha i} = +\frac{2}{a^2}K_{ik}K^{k}{}_{j}$.

Summing the four terms gives the stated result. □

Remark 3.1 (Independent check)**.** *This formula was independently confirmed by computing the full Riemann tensor of $g_{00} = a^2$, $g_{ij} = G_{ij}(t)$ by computer algebra (direct evaluation of $R^{\rho}{}_{\mu\rho\nu}$ from the Christoffel symbols, with $G_{ij}(t)$ left as a fully general symmetric 2×2 function of t), and by reduction to the one-dimensional case $ds^2 = dt^2 + b(t)^2 dx^2$, where R_{11} must equal the Gaussian curvature of the surface times g_{11}, i.e. $R_{11} = -\ddot{b}\, b = -b^2(\dot{H} + H^2)$ with $H = \dot{b}/b$; both checks reproduce Theorem* **??** *exactly and disagree with the original (unrevised) formula.*

These results show that in the purely time-dependent model, *all curvature arises from temporal deformation of the interaction metric.* The geometry is structurally analogous to a cosmological (FLRW-type) metric.

4 Stage II: State-Dependent Interaction Metric

4.1 Motivation for State Dependence

If $\gamma = \gamma(t)$ only, then $\partial_k G_{ij} = 0$ and the spatial geometry is flat. Many realistic interaction systems require the interaction rule to depend on the current interaction level:

- *Emotional escalation*: the coupling strength changes as the emotional intensity grows.
- *Biological saturation*: response functions saturate at high stimulation.
- *Adaptive feedback*: the system adjusts its own sensitivity based on state.

We therefore extend $\gamma = \gamma(t)$ to $\gamma = \gamma(t, I_S)$.

4.2 Extended Interaction Constraint

With $\gamma = \gamma(t, I_S)$, the differential of the constraint $I_P = \gamma I_S$ is

$$dI_P = \gamma\, dI_S + (\partial_t \gamma) I_S\, dt + (\partial_{I_S}\gamma)(dI_S)\, I_S.$$

At fixed time ($dt = 0$) we define the *interaction Jacobian*

$$J(t, I_S) = \gamma(t, I_S) + (\nabla_{I_S}\gamma(t, I_S))\, I_S, \quad \text{where} \quad (\nabla_{I_S}\gamma)_{ij} I_S^k = \frac{\partial \gamma_{ij}}{\partial I_S^k} I_S^k.$$

4.3 State-Dependent Metric

Substituting into the Euclidean line element gives the induced metric

$$G(t, I_S) = I + J(t, I_S)^T J(t, I_S).$$

This metric is symmetric, positive definite, and *depends on both t and I_S*. The full manifold metric remains block-diagonal:

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix}.$$

5 Stage II: Connection and Curvature with State Dependence

5.1 Levi-Civita Connection

Theorem 5.1 (Connection Structure)**.** *For the state-dependent metric, the non-vanishing Christoffel symbols are*

$$\begin{aligned}
\Gamma^0_{ij} &= -\frac{1}{2a^2}\partial_t G_{ij}, \\
\Gamma^i_{0j} &= K^i{}_j = \tfrac{1}{2} G^{ik}\partial_t G_{kj}, \\
\Gamma^i_{jk} &= \tfrac{1}{2} G^{il}\big(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}\big).
\end{aligned}$$

The spatial symbols Γ^i_{jk} are non-zero whenever $\nabla_{I_S}\gamma \neq 0$.

5.2 Ricci Tensor Decomposition

Theorem 5.2 (Ricci Tensor Decomposition)**.** *The Ricci tensor decomposes as*

$$R_{ij} = -\frac{1}{a^2}\left[\dot{K}_{ij} + \mathrm{Tr}(K)K_{ij} - 2K_{ik}K^k_j\right] + R^{\text{space}}_{ij},$$

where R^{space}_{ij} is the Ricci tensor of the spatial metric G_{ij} computed with respect to its own Levi-Civita connection.

Proof. Identical to the proof of Theorem 3.3, with two modifications now that $\Gamma^i_{jk} \neq 0$ in general. First, the second term $\partial_j \Gamma^\alpha_{i\alpha}$ now includes $\partial_j \Gamma^k_{ik}$, and the third and fourth terms acquire additional contributions from $\Gamma^{\alpha}_{\alpha\beta}\Gamma^{\beta}_{ij}$ and $\Gamma^{\alpha}_{j\beta}\Gamma^{\beta}_{\alpha i}$ with α, β both spatial. Because $g_{0i} = 0$ (the shift vector vanishes), no term mixes a single factor of $\Gamma^{0}_{\cdot\cdot}$ or $\Gamma^{i}_{0\cdot}$ with a single factor of Γ^{i}_{jk} inside R_{ij}; the temporal-type Christoffel symbols contract only among themselves, reproducing exactly the four terms computed in Theorem **??**, while the purely spatial Christoffel symbols contract only among themselves, reproducing exactly the Ricci tensor of the 2-dimensional Riemannian metric $G_{ij}(t, I_S)$ at fixed t. This is the standard Gauss–Codazzi splitting for a zero-shift, block-diagonal metric. Summing the two closed sets of terms gives the stated decomposition. □

Remark 5.1. *The decomposition cleanly separates two sources of curvature:*

1. Temporal curvature*: the $\dot{K}$ and K^2 terms arising from evolution of the interaction rule over time.*

2. Spatial curvature*: R^{space}_{ij} arising from the state-dependent variation $\partial_k G_{ij} \neq 0$.*

This geometric splitting is central to the interpretation of emotional escalation and non-linear feedback in the interaction manifold.

Theorem 5.3 (Temporal Component)**.**

$$R_{00} = -(\mathrm{Tr}(\dot{K}) + \mathrm{Tr}(K^2)).$$

5.3 Non-Triviality of the Interaction Geometry

Theorem 5.4 (Non-trivial Curvature Condition)**.** *If $\nabla_{I_S}\gamma \neq 0$ in some region of the interaction space, then the spatial curvature R^{space}_{ij} is generically non-zero.*

Proof. We have $\partial_k G_{ij} = (\partial_k J)^T J + J^T(\partial_k J)$. If $\nabla_{I_S}\gamma \neq 0$ then $\partial_k J \neq 0$, so $\partial_k G_{ij} \neq 0$. Non-vanishing spatial derivatives of the metric produce non-zero spatial Christoffel symbols, which in turn generate non-zero R^{space}_{ij}. □

The purely time-dependent model ($\gamma = \gamma(t)$) is recovered as the special case $\nabla_{I_S}\gamma = 0$, for which $R^{\text{space}}_{ij} = 0$.

6 Interaction Action Functional and Field Equations

6.1 Axioms of the Interaction Action

To derive a governing equation for the interaction geometry, we impose four physical axioms.

Axiom 6.1 (Locality). *The interaction dynamics at each point depend only on the metric and its derivatives up to finite order at that point:*

$$S[g] = \int_{\mathcal{M}} \mathcal{L}(g, \partial g, \partial^2 g, \ldots)\, d\mu_g.$$

Axiom 6.2 (Diffeomorphism Covariance). *The action is invariant under smooth coordinate transformations $x^\mu \mapsto x'^\mu(x)$:*

$$S[g] = S[\varphi^* g] \quad \forall\, \varphi \in \mathrm{Diff}(\mathcal{M}).$$

Axiom 6.3 (Metric Dynamics). *The metric $g_{\mu\nu}$ is the sole dynamical geometric field; the Lagrangian depends only on tensors generated from g.*

Axiom 6.4 (Second-Order Field Equations). *The Euler–Lagrange equations derived from the action contain at most second derivatives of the metric, ensuring well-posed interaction evolution.*

6.2 Derivation of the Unique Action

Lemma 6.1 (Covariant Scalar Construction). *Any diffeomorphism-invariant local Lagrangian must be constructed from contractions of the Riemann tensor $R^\rho{}_{\sigma\mu\nu}$ and its covariant derivatives.*

Lemma 6.2 (Second-Order Constraint). *If the Lagrangian contains ∇R or higher, the field equations contain derivatives of order ≥ 3 in the metric, violating Axiom 4.*

Proof. Since $R \sim \partial^2 g$, any $\nabla R \sim \partial^3 g$, so $\delta S \sim \partial^4 g$, giving fourth-order equations. □

Corollary 6.3. *The admissible Lagrangian depends only algebraically on the curvature: $\mathcal{L} = \mathcal{L}(g, R_{\mu\nu\rho\sigma})$.*

Lemma 6.4 (3D Curvature Identity). *In three spacetime dimensions, the Riemann tensor is completely determined by the Ricci tensor:*

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} - g_{\nu\rho}R_{\mu\sigma} + g_{\nu\sigma}R_{\mu\rho} - \frac{R}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

Every scalar invariant therefore reduces to a function of R and $R_{\mu\nu}R^{\mu\nu}$.

Lemma 6.5. *Quadratic curvature invariants such as $\int R_{\mu\nu}R^{\mu\nu}\sqrt{|g|}$ generate fourth-order field equations and are therefore excluded by Axiom 4.*

6.3 Main Theorem: Uniqueness of the Interaction Action

Theorem 6.6 (Uniqueness of the Interaction Action Functional)**.** *Let* $(\mathcal{M}, g)$ *be a three-dimensional interaction manifold with coordinates* (t, I_S^1, I_S^2) *equipped with the block-diagonal metric induced by the state-dependent interaction rule. Under Axioms 1–4, the unique admissible action functional is, up to a boundary term,*

$$S[g] = \int_{\mathcal{M}} \left(\frac{1}{2\kappa} R + \Lambda \right) \sqrt{|g|}\, d^3x,$$

where R *is the Ricci scalar,* κ *is a coupling constant, and* Λ *is a cosmological-type constant. Variation of this action yields the* interaction field equation

$$\boxed{R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu},}$$

where the interaction source tensor $T_{\mu\nu}$ *arises from matter or interaction contributions to be specified in applications.*

Proof. The argument proceeds in six steps:

1. **Locality and Covariance.** By Axiom 1–2 and the non-tensorial nature of Christoffel symbols, the Lagrangian must be built from the Riemann tensor and its covariant derivatives.

2. **Dimensional Reduction.** In three dimensions the Weyl tensor vanishes and every scalar invariant reduces to a function of R and $R_{\mu\nu}R^{\mu\nu}$.

3. **Second-Order Constraint.** By Axioms 4 and the preceding lemmas, all terms involving ∇R or quadratic curvature are excluded.

4. **Admissible Lagrangians.** The only surviving scalars are 1 (the volume element) and R (linear in curvature), giving $\mathcal{L} = \alpha + \beta R$.

5. **Normalization.** Setting $\beta = 1/(2\kappa)$ and $\alpha = \Lambda$ produces the stated action.

6. **Variation.** Standard variation of the Einstein–Hilbert term yields the Einstein tensor $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$; the cosmological term contributes $\Lambda g_{\mu\nu}$; and any interaction sources produce $\kappa T_{\mu\nu}$.

7. **Uniqueness.** Any other diffeomorphism-invariant local scalar leading to second-order equations must reduce to the above form in three dimensions.

□

The interaction field equation

$$R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu}$$

is structurally identical to the Einstein field equations of general relativity, but here it governs the geometry of the interaction space rather than spacetime.

6.4 Consistency with the Contracted Bianchi Identity

We now verify that the corrected Ricci tensor is compatible with the geometric identity that any genuine Ricci tensor must satisfy, and show that this consistency is precisely what fails for the original (erroneous) formula.

Lemma 6.7 (Second Bianchi Identity)**.** *For the Levi-Civita connection of any pseudo-Riemannian metric g,*

$$\nabla_\lambda R^\rho{}_{\sigma\mu\nu} + \nabla_\mu R^\rho{}_{\sigma\nu\lambda} + \nabla_\nu R^\rho{}_{\sigma\lambda\mu} = 0.$$

Proof. Fix a point p and choose Riemann normal coordinates centred at p, so that $\Gamma^\lambda_{\mu\nu}(p) = 0$ (though $\partial\Gamma$ need not vanish). At p, the covariant derivative of any tensor reduces to the ordinary partial derivative, since all connection terms in the covariant-derivative formula carry an explicit factor of $\Gamma(p) = 0$. Likewise, at p the Riemann tensor reduces to

$$R^\rho{}_{\sigma\mu\nu}\Big|_p = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma}\Big|_p,$$

because the quadratic $\Gamma\Gamma$ terms vanish at p. Hence at p,

$$\nabla_\lambda R^\rho{}_{\sigma\mu\nu}\Big|_p = \partial_\lambda\partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\lambda\partial_\nu \Gamma^\rho_{\mu\sigma}\Big|_p.$$

Summing the three cyclic permutations of (λ, μ, ν), every term of the form $\partial_a\partial_b\Gamma^\rho_{c\sigma}$ appears exactly twice with opposite sign (using $\partial_a\partial_b = \partial_b\partial_a$), so the sum vanishes identically at p. Since p was an arbitrary point and both sides of the claimed identity are tensorial (hence coordinate-independent statements once true in one coordinate system at a point), the identity holds everywhere. □

Corollary 6.8 (Contracted Bianchi Identity)**.** *For the Levi-Civita connection of any pseudo-Riemannian metric g,*

$$\nabla^\mu\Big(R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu}\Big) = 0.$$

Proof. Contract Lemma 6.7 on $\rho = \lambda$ (sum over ρ):

$$\nabla_\rho R^\rho{}_{\sigma\mu\nu} + \nabla_\mu R_{\sigma\nu} - \nabla_\nu R_{\sigma\mu} = 0,$$

where we used $R^\rho{}_{\sigma\nu\rho} = -R_{\sigma\nu}$ and $R^\rho{}_{\sigma\rho\mu} = R_{\sigma\mu}$. Contract again with $g^{\sigma\mu}$:

$$\nabla_\rho R^\rho{}_\nu + \nabla^\mu R_{\mu\nu} - \nabla_\nu R = 0 \implies 2\nabla^\mu R_{\mu\nu} = \nabla_\nu R.$$

Hence $\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R$, which is equivalent to $\nabla^\mu(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}) = 0$. □

Proposition 6.1 (Consistency of the Corrected Interaction Ricci Tensor)**.** *The Ricci tensor of Theorems 3.3 and 5.2, together with the temporal component* $R_{00} = -\mathrm{Tr}(\dot{K}) - \mathrm{Tr}(K^2)$ *of Theorem 3.2, satisfies the contracted Bianchi identity,*

$$\nabla^\mu\Big(R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu}\Big) = 0, \qquad \nu = 0, 1, 2.$$

Proof. By construction, R_{00} and the corrected R_{ij} of Theorems 3.3–5.2 are exactly the components $R^\rho{}_{\mu\rho\nu}$ obtained by contracting the Riemann tensor of the Levi-Civita connection of g given in Theorem 3.1/5.1 (this is what the proofs of those theorems compute: each is derived directly from the same connection, term by term, with no independent postulation). They are therefore the genuine Ricci tensor of (M, g) in the sense required by Lemma 6.7, and Corollary 6.8 applies directly. ■ □

Proposition 6.2 (Violation by the Original Formula)**.** *Let* $\widetilde{R}_{ij} := \dot{K}_{ij} + K_{ik}K^k{}_j + K_{ij}\mathrm{Tr}(K)$ *denote the spatial tensor originally stated in Theorem 3.3, and let* $\widetilde{G}_{\mu\nu} := \widetilde{R}_{\mu\nu} - \frac{1}{2}\widetilde{R}\,g_{\mu\nu}$ *be the associated Einstein-type tensor (with* R_{00} *unchanged). Then, generically,* $\nabla^\mu\widetilde{G}_{\mu 0} \neq 0$.

Proof sketch. $\widetilde{R}_{ij}$ differs from the true Ricci tensor R_{ij} of Theorem **??** by

$$\Delta_{ij} := \widetilde{R}_{ij} - R_{ij} = \big(1 + \tfrac{1}{a^2}\big)\dot{K}_{ij} + \big(1 + \tfrac{1}{a^2}\big)\mathrm{Tr}(K)K_{ij} - \big(1 - \tfrac{2}{a^2}\big)K_{ik}K^k{}_j,$$

a tensor that is *not* the Ricci tensor of any metric (it does not arise as a Riemann contraction of the connection Γ of Theorem 3.1). Since Corollary 6.8 is a property specific to genuine curvature tensors of a fixed connection, and since covariant divergence is linear, $\nabla^\mu\widetilde{G}_{\mu\nu} = \nabla^\mu G_{\mu\nu} + (\text{divergence of the } \Delta\text{–induced terms}) = 0 + (\text{divergence of } \Delta\text{–terms})$, there is no a priori reason for the second piece to vanish, and for a generic choice of $G_{ij}(t)$ it does not. Direct symbolic evaluation (computer algebra, Christoffel symbols of $g_{00} = a^2, g_{ij} = G_{ij}(t)$ applied to $\widetilde{G}_{\mu\nu}$) confirms $\nabla^\mu\widetilde{G}_{\mu 0} \neq 0$ for generic $G_{ij}(t)$, while $\nabla^\mu\widetilde{G}_{\mu i} = 0$ accidentally survives in this class of examples. This confirms concretely that the original Theorem 3.3 does not describe a genuine Ricci tensor. □

Remark 6.1 (Implication for the interaction field equation)**.** *Proposition 6.1 shows that, once Theorem 3.3/5.2 are replaced by their corrected forms, the interaction field equation of Theorem 6.6,*

$$R_{\mu\nu} - \tfrac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

automatically enforces the local conservation law $\nabla^\mu T_{\mu\nu} = 0$ for any interaction source tensor $T_{\mu\nu}$ substituted on the right-hand side, exactly as in general relativity. Under the original, uncorrected Theorem 3.3, this conservation law failed to be geometrically guaranteed, and had to be imposed as a separate, unmotivated constraint on $T_{\mu\nu}$ – which is the same class of pathology (loss of a geometrically protected conservation law) previously identified in earlier revisions of this series that led to withdrawal of the Einstein-field-equation framework. The present correction isolates that failure to a single, identifiable computational error in Theorem 3.3, rather than a structural defect of the action-based derivation in Section 6, which remains valid unchanged.

7 Physical Interpretation

7.1 Curvature as Interaction Geometry

The two-stage framework provides a geometric dictionary:

Geometric object	**Interaction interpretation**
Metric $G_{ij}(t, I_S)$	Structure of the interaction space
$K^i{}_j$	Rate of change of interaction coupling
E_{ij}	Intrinsic interaction acceleration
R_{00}	Negative trace of total interaction acceleration
R_{ij}	Spatial interaction curvature
$T_{\mu\nu}$	Reinforcement and perceptual sources
Λ	Baseline interaction intensity

7.2 Role of State Dependence

State dependence $\nabla_{I_S}\gamma \neq 0$ is essential for three reasons:

1. **Physical realism.** Real interaction systems exhibit state-dependent coupling (emotional escalation, biological saturation, adaptive feedback).
2. **Mathematical completeness.** Without state dependence the spatial geometry is flat ($R_{ij}^{\text{space}} = 0$) and the model reduces to Stage I.
3. **Field equation structure.** Non-trivial spatial curvature is necessary for the interaction field equation to have meaningful non-flat solutions.

8 Conclusion

This paper has developed a unified geometric formulation of two-agent interaction dynamics in two stages, culminating in an Einstein-type interaction field equation derived from first principles.

In **Stage I**, we constructed a time-dependent Riemannian metric from a linear interaction constraint $I_P = \gamma(t) I_S$. The interaction flow tensor $K^i{}_j$ and the emotional quantity tensor E_{ij} emerge naturally as geometric quantities governing Ricci curvature. All curvature in this stage arises from temporal deformation, analogous to cosmological metric evolution.

In **Stage II**, we generalised to $\gamma = \gamma(t, I_S)$, introducing spatial variation in the metric and non-trivial intrinsic curvature. The Ricci tensor decomposes cleanly into temporal and spatial contributions, with the spatial term generated by state dependence of the interaction rule.

Finally, we established a **variational principle**: under the axioms of locality, covariance, and second-order dynamics, the interaction geometry is uniquely governed by

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = \kappa\, T_{\mu\nu}.$$

This equation provides a rigorous bridge between differential geometry and behavioral science, with $T_{\mu\nu}$ encoding reinforcement effects and perceptual forcing.

Future work will focus on:

- extending the manifold to higher-dimensional interaction spaces,
- specifying explicit forms of the interaction source tensor $T_{\mu\nu}$ for cognitive and social systems,
- exploring empirical validation within behavioral and neuroscientific datasets,
- investigating connections to matrix Ricci flow and optimal transport.

Acknowledgements

The author thanks the mathematical physics and behavioral science communities for foundational insights that motivated this work.

References

[1] S. I. Amari. *Information Geometry and Its Applications.* Springer, 2016.

[2] K. J. Friston et al. The free energy principle and the geometry of agency. *Journal of Mathematical Psychology*, 2023.

[3] M. P. Hobson et al. *General Relativity: An Introduction for Physicists*. Cambridge University Press, 2006.

[4] N. D. Volkow et al. The dopamine motive system: Implications for drug addiction. *Lancet Psychiatry*, 10(4), 2023.